

Spontaneous Intracerebral Hemorrhage

AHA/ASA 2022 Guidelines for acute management of spontaneous ICH.

1. Blood Pressure Management

- **Target:** For presenting SBP 150-220 mmHg, target **140 mmHg** (range 130-150 mmHg).
- **Timing:** Initiate early (within 2 hours) and maintain **sustained control**.
- **Variability:** Avoid large swings in BP; smooth titration is associated with better outcomes.
- **Agents:** Nicardipine, Clevidipine, or Labetalol infusions preferred.

2. Anticoagulation Reversal

- **VKA (Warfarin):** 4-Factor Prothrombin Complex Concentrate (PCC) + Vitamin K (10mg IV).
- **Dabigatran:** Idarucizumab (5g IV) or PCC/aPCC if unavailable.
- **Factor Xa Inhibitors (Apixaban, Rivaroxaban):** Andexanet alfa or 4-Factor PCC.
- **Heparin (UFH/LMWH):** Protamine sulfate.

3. Surgical Management

- **Cerebellar ICH:** Urgent evacuation indicated for diameter > 3 cm, brainstem compression, or hydrocephalus.
- **Supratentorial:** Life-saving decompression (craniectomy) for deteriorating patients with large lobar ICH.
- **Routine Evacuation:** STICH II showed no overall benefit for early routine evacuation of supratentorial ICH.
- **Minimally Invasive Surgery:** MISTIE III showed no functional benefit but improved hematoma clearance; may be considered in select cases.

4. Medical & Supportive Care

- **EVD Placement:** For GCS $\leq$ 8, significant IVH, or hydrocephalus causing clinical decline.
- **Seizures:** Treat clinical seizures; prophylactic antiseizure medications are **not** recommended.
- **DVT Prophylaxis:** Pneumatic compression initially. Pharmacologic prophylaxis (LMWH/UFH) typically starting 24-48h post-onset after stability is confirmed on repeat imaging.

AHA/ASA 2022: Greenberg SM et al. *Stroke*. 2022;53:e282-e361. PMID: 35579034 | **STICH II:** Mendelow AD et al. *Lancet*. 2013;382:397-408. PMID: 23726393 | **MISTIE III:** Hanley DF et al. *Lancet*. 2019;393:1021-1032. PMID: 30739747